

Learning Objectives

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Learning Objective 1. Describe hemodynamics. Explain the relationship between **blood flow**, **velocity**, **pressure**, **vascular resistance** and **cross-sectional area** of blood vessel.

Learning Objective 2. Describe blood pressure and its components, including cardiac output and total peripheral resistance

Learning Objective 3. Predict factors that can alter cardiac output, stroke volume and pulse pressure.

Learning Objective 4. Evaluate and determine cardiac output given laboratory and physical findings.

Learning Objective 5. Describe Poiseuille's law and its relationship with **blood flow** and describe the La Place Law and its relevance to the cardiovascular system.

Hemodynamics

Hemodynamics refers to the principles that govern blood flow in the cardiovascular system. It refers to the principles that drives blood flow in the cardiovascular system. Velocity, Pressure, Cross-Sectional Area, Resistance and **capacitance or compliance** (capacity to hold blood).

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Fluid Movement

Fluid Dynamics

What is fluid?

Examples:

Water



oil



Blood



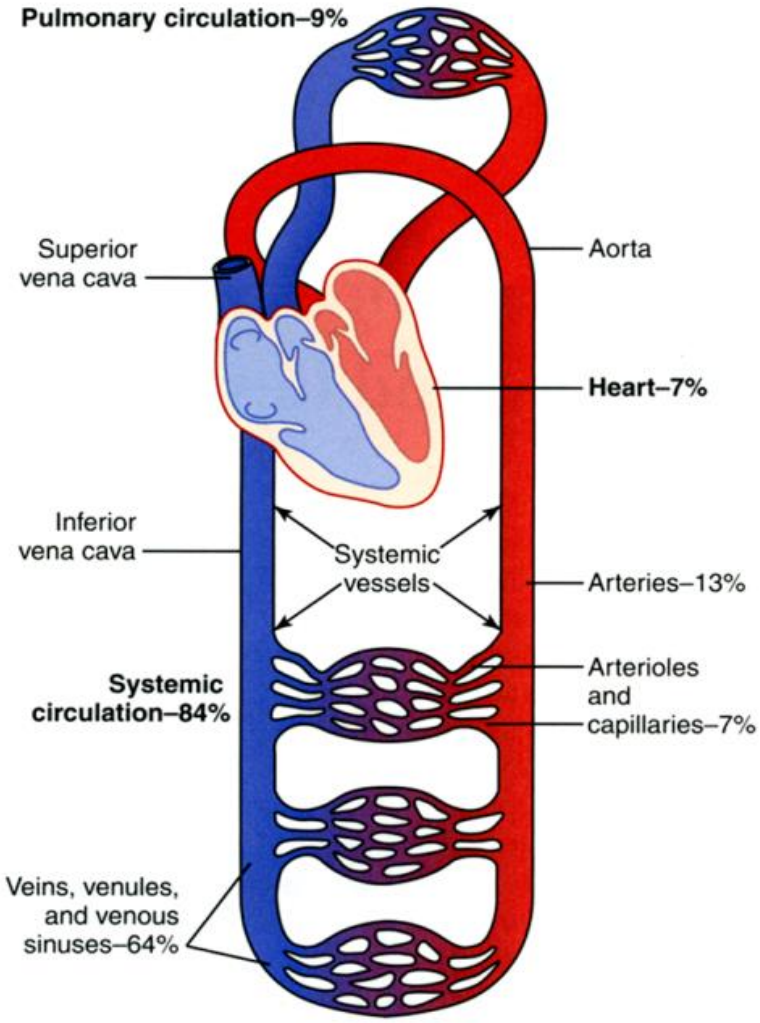
Honey



Physical Characteristics of Circulation

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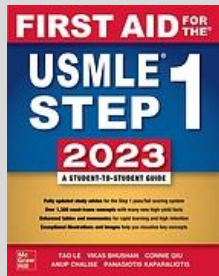
- 1) Set of closed vessels. **“Network of pipes”**
- 2) Pressure Pump (heart).
- 3) Circulating fluid (blood).

♥ **Stroke Volume (SV)** is the volume of blood pumped out of the heart with each beat

♥ **Cardiac Output (CO)** is volume of blood pumped by the heart per minute.

$CO = \text{stroke volume (mL)} \times \text{heart rate (beats/min)}$

♥ Cardiac Output is ~5 L/min.



Basic Terminology in Circulation

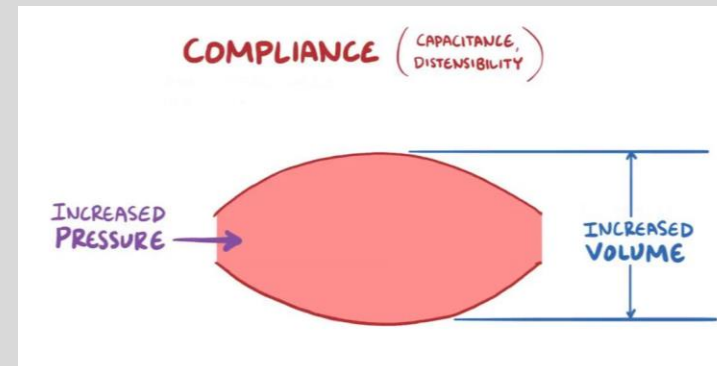
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Blood Flow is the volume of blood that passes a given point in the circulation in a given period of time (mL/sec or cm^3/sec). **volume/time**

Velocity measures the distance moved in a given amount of time. It is the distance of displacement of blood per unit time (cm/sec). **distance/time**

Compliance or **Capacitance** or **Distensibility** is the ability of a vessel to distend and increase volume with increasing transmural pressure. How much a vessel can increase the volume of blood, how much it can hold.

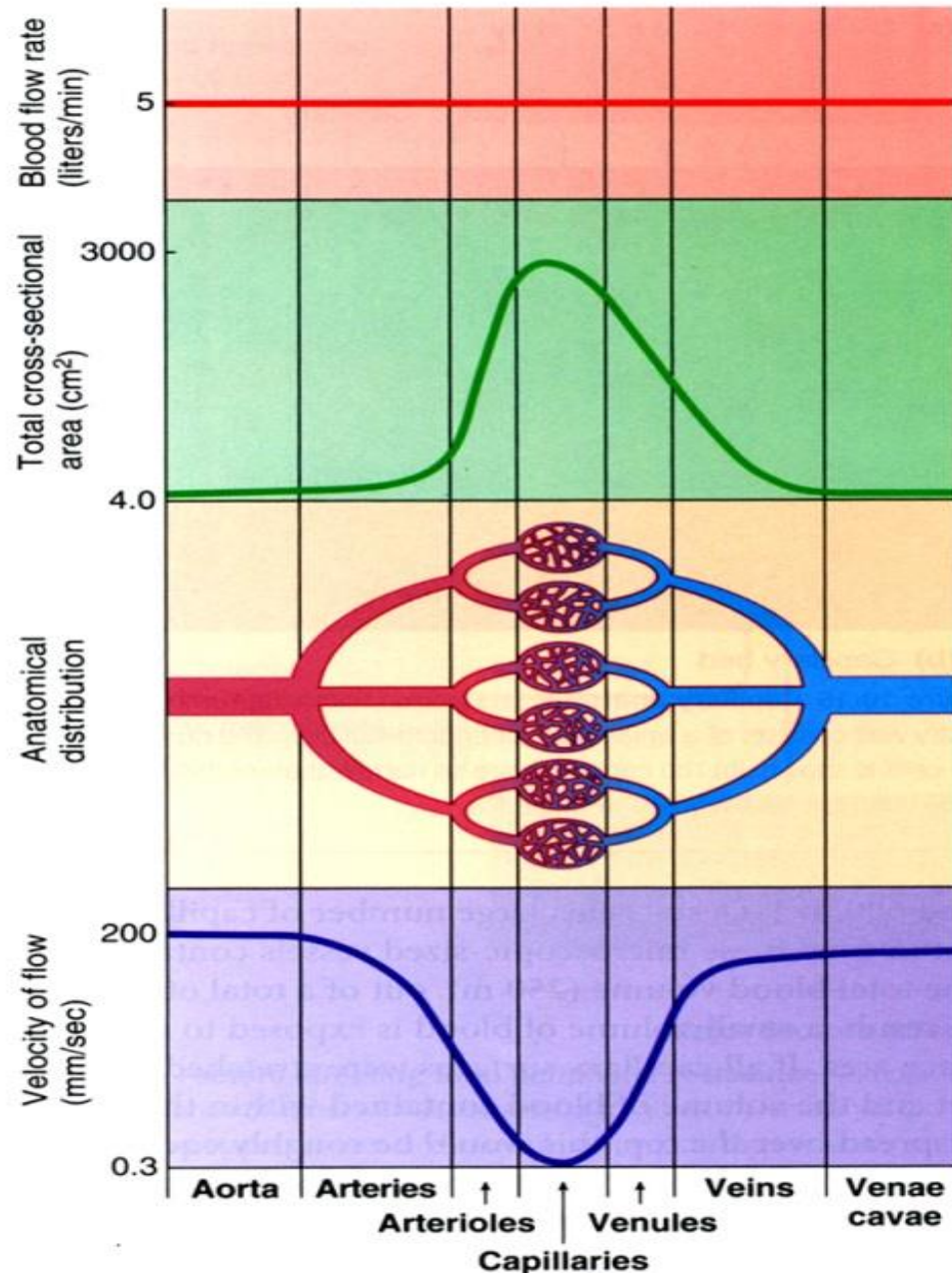


Vascular Cross-Sectional Area and Blood Flow

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$$\text{Velocity} = \frac{Q}{A} = \frac{Q}{\pi r^2}$$



♥ Flow rate remains **constant** because velocity changes with the changes in vessel radius.

♥ Capillaries have highest total cross-sectional area and lowest blood velocity

♥ *Velocity of blood flow is highest in the aorta and lowest in the capillaries because of a **dramatic increase in total cross-sectional area**. Velocity increases again as blood moves from the capillaries to the venules to veins, and finally, to the vena cava because total cross-sectional area is decreasing again.*

Practice Question

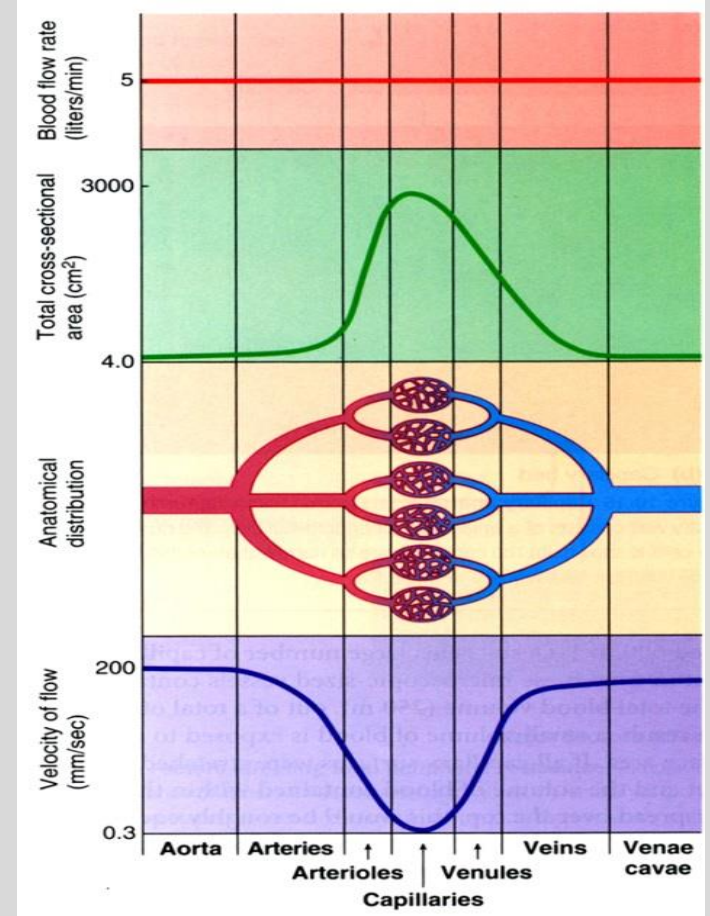
A man has a cardiac output of 5.5L/min. The diameter of his thoracic aorta is estimated to be 20mm, and the total cross-sectional area of his systemic capillaries is estimated to be 2500 cm². What is the velocity of blood flow in the aorta relative to the velocity of blood flow in the capillaries?

Capillaries: Velocity = $Q/A = 5.5\text{L/min} / 25000 = 2.2 \text{ cm/min}$

$$\text{Area} = \pi r^2$$

$$\text{Area} = 3.14 \times (10\text{mm})^2 = 3.14 \times (1\text{cm})^2 = 3.14 \text{ cm}^2$$

Aorta: Velocity = $Q/A = 5.5\text{L/min} / 3.14 = 1752\text{cm/min}$



Summary:

To keep **constant blood flow**, we need to take into consideration: cross section area of the vessels, and velocity

♥ Velocity ✓

Next:

- ♥ Pressure
- ♥ Resistance

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$$\text{Velocity} = \frac{Q}{A} = \frac{Q}{\pi r^2}$$

Blood Flow: Pressure and Resistance

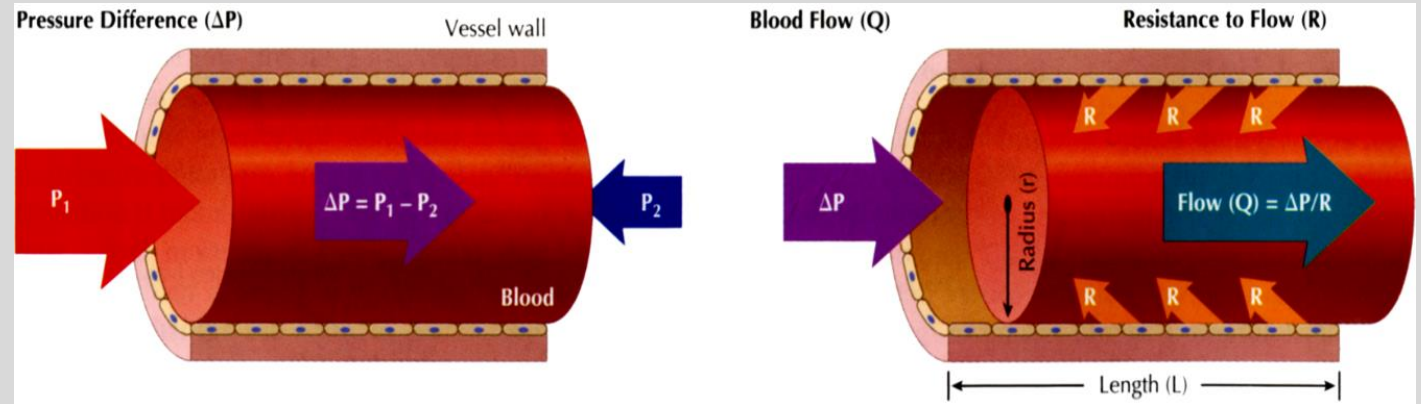
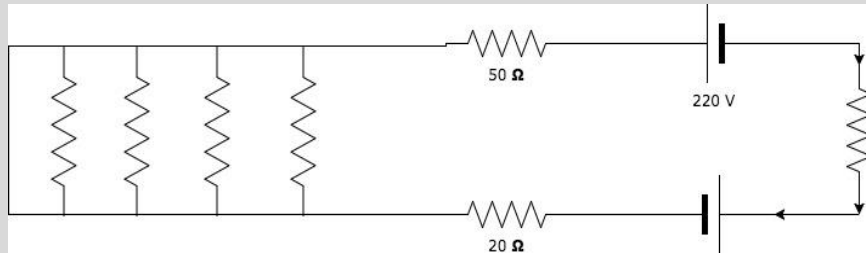
Ohm's Law, fundamental law of physics, states that current (I) equals the voltage difference (ΔV) divided by resistance (R). In relating Ohm's Law to fluid flow, the voltage difference is the **pressure difference** (ΔP ; sometimes called **driving pressure**, **perfusion pressure**, or **pressure gradient**), the resistance is the resistance to flow (R) offered by the blood vessel and its interactions with the flowing blood, and the current is the blood flow (Q).

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Electrical Circuit - Ohm's Law : $I = \frac{\Delta \text{Voltage}}{\text{Resistance}}$

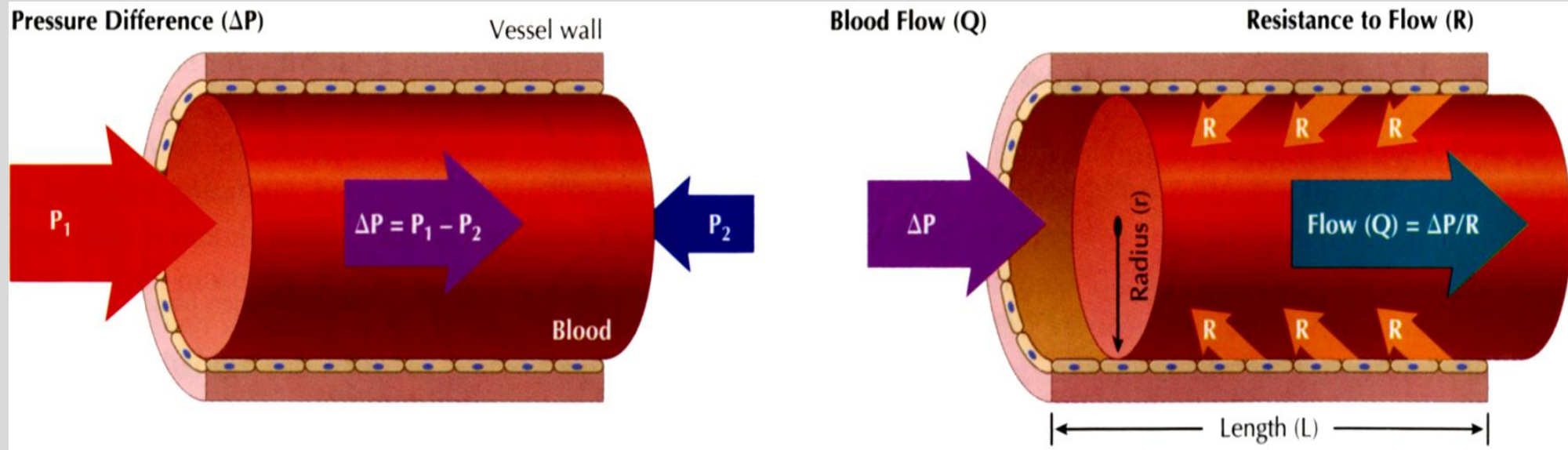
Blood flow (Q): $Q = \frac{P_1 - P_2}{R} = \frac{\Delta P}{R}$ (mL/min)



The major mechanism for changing blood flow in the cardiovascular system is by changing the resistance of blood vessels, particularly the arterioles

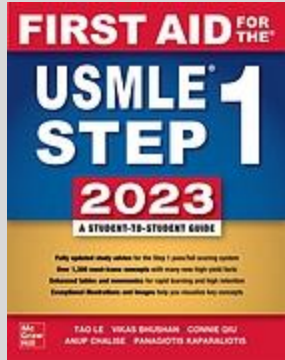
Blood Flow, Pressure and Resistance

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Pressure Difference (ΔP) is estimated as pressure difference of the blood between the two ends of the vessel segment: $P_1 - P_2 = \Delta P$ (mmHg). Flow is always from higher(P_1) to lower(P_2) pressure, from arteries to veins.

Vascular resistance (R) to blood flow occurs as a *result of friction between the flowing blood and the intravascular endothelium* along the inside of the vessel.



Practice Question

A 55-year-old-woman has renal blood flow of 500mL/min. Her renal arterial pressure is 100 mmHg and renal venous pressure is 10 mmHg. What is the vascular resistance of the left kidney in this woman?

$$Q = \frac{\Delta P}{R}$$

$$500 = \frac{100 - 10}{R}$$

$$R = \frac{90}{500} = 0.18 \text{ mmHg/ml/min}$$

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Next:

- ♥ Velocity ✓
- ♥ Pressure ✓
- ♥ Resistance

The relationship between resistance, blood vessels diameter (radius) and blood viscosity is described by the Poiseuille Equation

Practice Question

A 65-year-old man with peptic ulcer disease comes to the emergency department due to nausea, melena and feels faint. His blood pressure is 90/40 mmHg, pulse is 130/min, and respirations are 28/min. Physical examination reveals a lethargic man with cold extremities. The patient receives blood transfusion; however, the transfusion rate is very slow, and the patient remains hypotensive. Which of the following will lead to the greatest improvement in flow rate?

- a. Doubling the diameter of the intravenous catheter
- b. Doubling the length of the intravenous catheter
- c. Doubling the pressure of the infusion pump
- d. Placing a second intravenous catheter with identical diameter and length

Poiseuille's Equation

$$\text{Flow (Q)} = \frac{\Delta P \pi r^4}{8L\eta}$$

Hypovolemic Hemorrhagic Shock

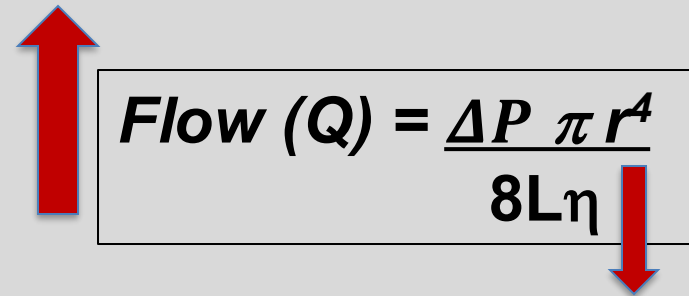
Practice Question

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A 22-year-old medical student presents with a 3-month history of fatigue and complain of feeling sleepy most of the time. She is found to have a mid-systolic murmur and higher than normal cardiac output. The differential diagnosis based on these findings includes which of the following?

- a.) Cardiac tamponade
- b.) Mitral regurgitation
- c.) Third-degree heart block
- d.) Anemia



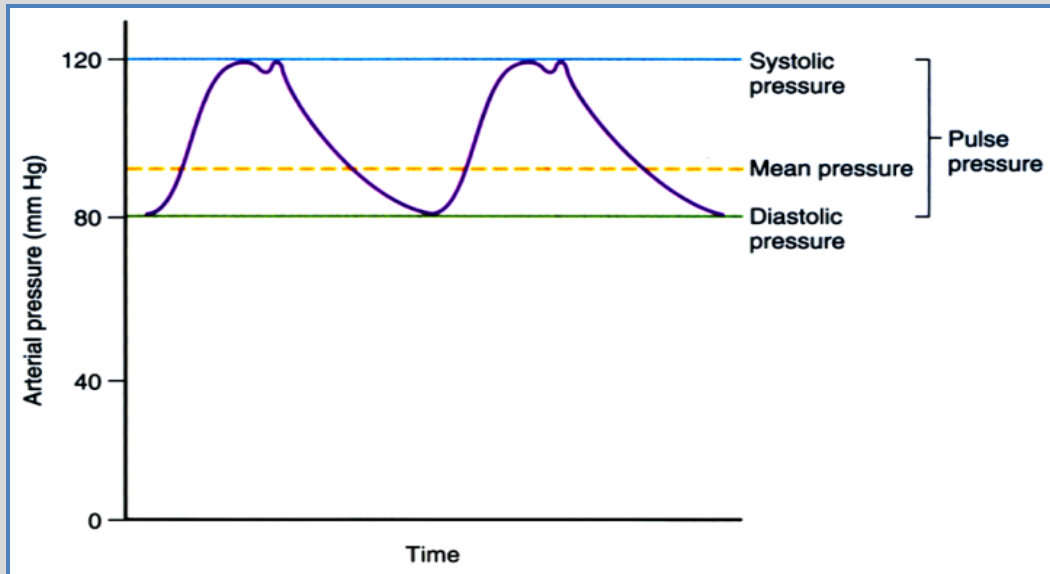
The diagram shows the equation for flow (Q) in a tube, known as Poiseuille's Law. The equation is enclosed in a black rectangular box. A large red arrow points upwards from the bottom left of the box, and another large red arrow points downwards from the bottom right of the box, indicating the direction of flow through the tube.

$$Flow (Q) = \frac{\Delta P \pi r^4}{8L\eta}$$

Systemic Arterial Pressures during the Cardiac Cycle

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$$\text{Mean Arterial Pressure} = \text{Diastolic Pressure} + \left(\frac{1}{3} \times \text{Pulse Pressure} \right)$$

$$\text{Pulse Pressure} = \text{Systolic Pressure} - \text{Diastolic Pressure}$$

- ♥ **Systolic Pressure** is the *highest arterial pressure* measured during a cardiac cycle.
 - ♥ **Diastolic Pressure** is the *lowest arterial pressure* measured during a cardiac cycle.
 - ♥ **Mean Arterial Pressure** is the running average of arterial pressure in the body
 - ♥ **Pulse Pressure** is the *difference between systolic pressure and diastolic pressure*.
- The *magnitude of the pulse pressure reflects the volume of blood ejected on a single beat*.

Definitions

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For the entire systemic circulation:

Pulse Pressure: Systolic Pressure – Diastolic Pressure

Blood Pressure (BP): Cardiac Output x Total Peripheral Resistance

Total Peripheral Resistance (TPR) = Systemic Vascular Resistance (SVR)

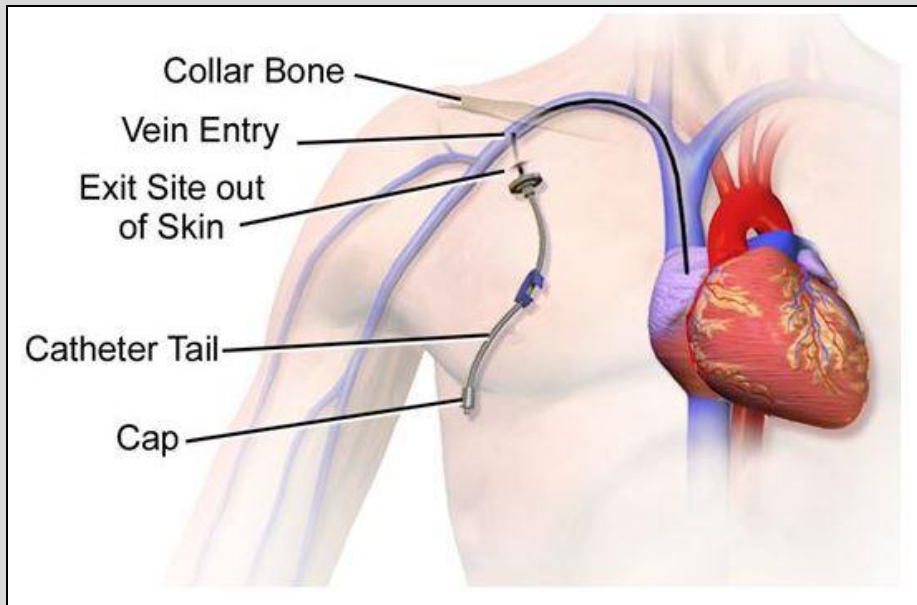
Mean Arterial Pressure (MAP): $\frac{2 \times \text{Diastolic Pressure} + \text{Systolic Pressure}}{3}$

Cardiac Output: $\frac{\text{Aorta Pressure} - \text{Vena Cava Pressure}}{\text{Total Peripheral Resistance}}$

Central Venous Pressure (CVP)

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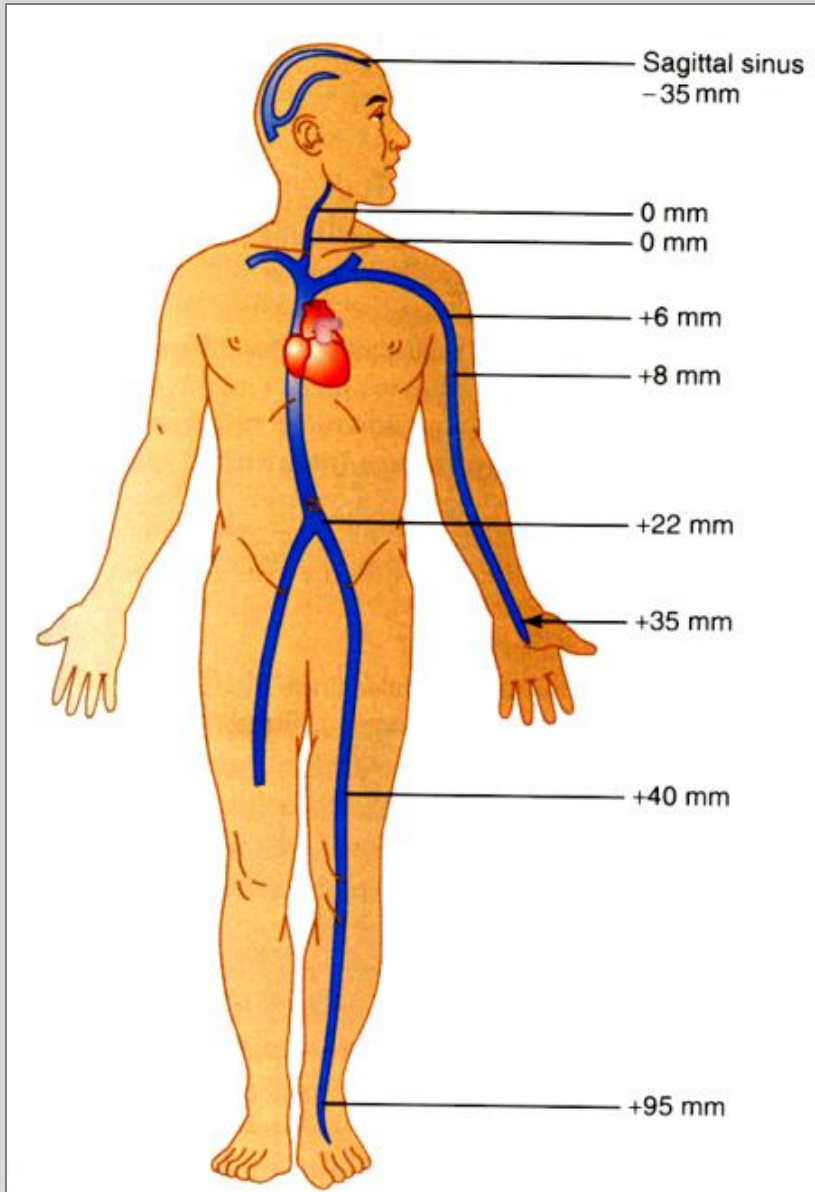
Blood pressure in thoracic vena cava, near right atrium

It is regulated by a balance between:

- the ability of the heart to pump blood out of the right atrium and ventricle into the lungs;
- the tendency for the blood to flow from the peripheral veins into the right atrium (venous return).

Normal Central Venous Pressure (CVP): 0-7 mmHg

- Weight of the blood in vessels is affected by gravitational force that generates hydrostatic pressure within the vascular system.



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- In a *standing adult*, *central venous pressure in the right atrium is ~0 mm Hg*, while *hydrostatic pressure in the veins of the feet can reach ~95 mm Hg* (depends on a person's height) *because of the weight of the blood in the veins* between the heart and the feet ($130 \text{ cm} / 13.6 \text{ mm} \approx 95 \text{ mm Hg}$).
- Gravitational venous *pressures at other levels* of the body, except neck and head, are *between ~0 and ~95 mm Hg*.
- If* there were *no one-way valves in veins*, the *hydrostatic pressure in a standing adult* would cause the venous pressure in the feet always to be *about 95 mm Hg*. However, under *ordinary circumstances*, the *venous pressure in the feet* remains *less than 30 mm Hg* because of the *leg skeletal muscle pump*.

Practice Question

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A 27-year-old medical student is sitting in silent study reviewing cardiovascular physiology. She takes her pulse and counts her heart rate to be 75 beats/min. Her friend preparing for ICC measures her blood pressure and tells her she has a systolic pressure of 110 mmHg and a diastolic pressure of 70 mmHg. She knows her average stroke volume is 65mL/beat. Based on this information:

What is her Cardiac Output?

What is her Mean Arterial Pressure?

What is her Pulse Pressure?

Summary Question:

A 64 year old female presents to the office following a hospital visit for diabetic ketoacidosis. Her medical reports show that at the time of admission the patient had the following vitals signs: Heart rate 55 beats/min a respiratory rate of 27 breaths/min systolic blood pressure was 156 mmHg and diastolic pressure was 85 mmHg. Echocardiogram revealed a right atrial pressure of 1 mmHg. A venous Doppler revealed poor upper and lower extremity perfusion. Arterial blood gas results showed an arterial O_2 concentration of 24mL O_2 /100mL. Venous blood gas results showed a venous O_2 concentration of 18 mL O_2 /100mL. Metabolic oxygen consumption rate was determined to be 150mL/min. Based on these reports what was the patients systemic vascular resistance at the time of admission?

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